

Engineering Chemistry (ENSH 153)

Complete Solved Question Bank

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Chapter 1

Electrochemistry

Question 2083 Baishakh Q1

Problem: Define salt bridge. What is the role of salt bridge? From the given information, answer the following questions. $\text{Ni} \longrightarrow \text{Ni}^{2+} + 2\text{e}^- \quad E^\circ = 0.24 \text{ V}$
 $\text{Cd} \longrightarrow \text{Cd}^{2+} + 2\text{e}^- \quad E^\circ = 0.40 \text{ V}$ $[\text{Ni}^{2+}] = 0.1 \text{ M}$, $[\text{Cd}^{2+}] = 0.01 \text{ M}$ i) Identify anode and cathode while constructing the galvanic cell with reasons. ii) Write the symbolic representation of the galvanic cell when above given electrodes are coupled. iii) Calculate the EMF of the galvanic cell at 25°C .

Solution: A **salt bridge** is a U-shaped tube containing an inert electrolyte (such as KCl, KNO_3 , or NH_4NO_3) in an agar-agar gel. Its primary roles are:

- To complete the electrical circuit by allowing the flow of ions between the two half-cells.
- To maintain electrical neutrality in both half-cells by providing counter-ions to balance the charge generated during the electrochemical reactions.

Given oxidation potentials: $E^\circ_{\text{Ni}/\text{Ni}^{2+}} = 0.24 \text{ V}$ $E^\circ_{\text{Cd}/\text{Cd}^{2+}} = 0.40 \text{ V}$

i) Anode and Cathode: The electrode with the higher oxidation potential acts as the anode, and the one with the lower oxidation potential acts as the cathode. Since $E^\circ_{\text{Cd}/\text{Cd}^{2+}} > E^\circ_{\text{Ni}/\text{Ni}^{2+}}$, Cadmium (Cd) acts as the **anode** (oxidation occurs here), and Nickel (Ni) acts as the **cathode** (reduction occurs here).

ii) Symbolic representation: At anode: $\text{Cd}_{(\text{s})} \mid \text{Cd}^{2+}_{(\text{aq})} (0.01 \text{ M})$ At cathode: $\text{Ni}^{2+}_{(\text{aq})} (0.1 \text{ M}) \mid \text{Ni}_{(\text{s})}$
Cell notation: $\text{Cd}_{(\text{s})} \mid \text{Cd}^{2+}_{(\text{aq})} (0.01 \text{ M}) \parallel \text{Ni}^{2+}_{(\text{aq})} (0.1 \text{ M}) \mid \text{Ni}_{(\text{s})}$

iii) EMF of the galvanic cell: Standard cell potential (E°_{cell}) is:

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}(\text{red})} - E^\circ_{\text{anode}(\text{red})}$$
$$E^\circ_{\text{cell}} = (-0.24 \text{ V}) - (-0.40 \text{ V}) = 0.16 \text{ V}$$

According to the Nernst equation at 25°C:

$$\begin{aligned}E_{\text{cell}} &= E_{\text{cell}}^{\circ} - \frac{0.0591}{n} \log \frac{[\text{Cd}^{2+}]}{[\text{Ni}^{2+}]} \\E_{\text{cell}} &= 0.16 - \frac{0.0591}{2} \log \frac{0.01}{0.1} \\E_{\text{cell}} &= 0.16 - 0.02955 \times \log(0.1) \\E_{\text{cell}} &= 0.16 - 0.02955 \times (-1) = 0.16 + 0.02955 = 0.18955 \text{ V}\end{aligned}$$

$E_{\text{cell}} = 0.1896 \text{ V}$

Question 2083 Baishakh Q1 OR

Problem: What is polarization and overpotential? Write Tafel plots equation and mention its applications.

Solution: Polarization: It is the deviation of the electrode potential from its equilibrium (reversible) value when a current flows through the cell. It generally decreases the efficiency of galvanic cells and increases the energy required for electrolysis.

Overpotential (Overvoltage): It is the extra potential that must be applied beyond the theoretical (reversible) potential to carry out an electrochemical reaction at a practical rate. It is mathematically the difference between the actual applied potential and the equilibrium potential: $\eta = E_{\text{applied}} - E_{\text{equilibrium}}$.

Tafel Equation: The relationship between overpotential (η) and current density (i) is given by the Tafel equation:

$$\eta = a \pm b \log i$$

where a and b are Tafel constants specific to the electrode material and the reaction, and i is the current density. The positive sign applies to anodic polarization and the negative sign to cathodic polarization.

Applications of Tafel Plots:

- Determination of the corrosion rate of metals.
- Estimation of the exchange current density (i_0) and the transfer coefficient.
- Evaluation of the efficiency and kinetics of electrocatalysts (e.g., in fuel cells and electrolyzers).

Question 2082 Bhadra Q1

Problem: What is a galvanic cell? Draw a diagram of a Daniel cell. Calculate the emf of the following cell at 25°C: $\text{Fe}/\text{Fe}^{2+}(0.6 \text{ M})//\text{Sn}^{2+}(0.2 \text{ M})/\text{Sn}$. Given: $E_{\text{Fe}^{2+}/\text{Fe}}^{\circ} = -0.44 \text{ V}$ and $E_{\text{Sn}^{2+}/\text{Sn}}^{\circ} = -0.14 \text{ V}$.

Solution: Galvanic Cell: A galvanic (or voltaic) cell is an electrochemical device that converts chemical energy from a spontaneous redox reaction into electrical energy.

Daniel Cell Diagram: (A diagram typically shows a Zinc anode in ZnSO_4 solution and a Copper cathode in CuSO_4 solution, connected by a salt bridge and an external circuit.)

Note: In an actual exam, draw two beakers, a U-tube salt bridge, an external wire with a voltmeter, labeling the Zn anode (negative) and Cu cathode (positive).

EMF Calculation: Given: $E^\circ_{\text{cathode}}(E^\circ_{\text{Sn}^{2+}/\text{Sn}}) = -0.14 \text{ V}$ $E^\circ_{\text{anode}}(E^\circ_{\text{Fe}^{2+}/\text{Fe}}) = -0.44 \text{ V}$

Standard cell potential:

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

$$E^\circ_{\text{cell}} = -0.14 - (-0.44) = 0.30 \text{ V}$$

Cell reaction: $\text{Fe}_{(\text{s})} + \text{Sn}_{(\text{aq})}^{2+} \longrightarrow \text{Fe}_{(\text{aq})}^{2+} + \text{Sn}_{(\text{s})}$ Here, $n = 2$. Using the Nernst equation:

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{2} \log \frac{[\text{Fe}^{2+}]}{[\text{Sn}^{2+}]}$$

$$E_{\text{cell}} = 0.30 - 0.02955 \log \left(\frac{0.6}{0.2} \right)$$

$$E_{\text{cell}} = 0.30 - 0.02955 \log(3)$$

$$E_{\text{cell}} = 0.30 - 0.02955 \times 0.4771$$

$$E_{\text{cell}} = 0.30 - 0.0141 = 0.2859 \text{ V}$$

$E_{\text{cell}} = 0.2859 \text{ V}$

Question 2082 Bhadra Q1 OR

Problem: Write down the Butler-Volmer equation and its uses. Explain the charging and discharging process in lead acid battery with suitable chemical reactions.

Solution: Butler-Volmer Equation: It describes how the electrical current through an electrode depends on the voltage difference across the electrode-electrolyte interface. The equation is:

$$i = i_0 \left[\exp \left(\frac{\alpha_a z F \eta}{RT} \right) - \exp \left(-\frac{\alpha_c z F \eta}{RT} \right) \right]$$

Where: i = electrode current density, i_0 = exchange current density, η = overpotential, α_a, α_c = anodic and cathodic charge transfer coefficients, z = number of electrons involved, F = Faraday's constant.

Uses: It is the fundamental equation in electrochemical kinetics used to study the rates of electron transfer, fuel cell design, battery modeling, and corrosion analysis.

Lead-Acid Battery Processes: A lead-acid battery consists of a Pb anode and a PbO_2 cathode immersed in H_2SO_4 electrolyte.

Discharging (Acting as Galvanic Cell): At Anode (Oxidation): $\text{Pb}_{(\text{s})} + \text{SO}_4^{2-}(\text{aq}) \longrightarrow \text{PbSO}_4(\text{s}) + 2\text{e}^-$ At Cathode (Reduction): $\text{PbO}_2(\text{s}) + 4\text{H}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) + 2\text{e}^- \longrightarrow \text{PbSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l})$ Overall Discharging Reaction: $\text{Pb}_{(\text{s})} + \text{PbO}_2(\text{s}) + 2\text{H}_2\text{SO}_4(\text{aq}) \longrightarrow 2\text{PbSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l})$

Charging (Acting as Electrolytic Cell): An external voltage is applied, reversing the reactions. Overall Charging Reaction: $2\text{PbSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \longrightarrow \text{Pb}(\text{s}) + \text{PbO}_2(\text{s}) + 2\text{H}_2\text{SO}_4(\text{aq})$

Question 2082 Baishakh Q1

Problem: Define polarization and overvoltage.

Solution: Polarization: Polarization is the shift or deviation of the electrode potential from its equilibrium value upon the passage of an electric current. It is caused by a slow step in the electrochemical reaction sequence (such as mass transport or charge transfer) and reduces the operational efficiency of electrochemical cells.

Overvoltage (Overpotential): It is the quantifiable difference between the actual potential required for an electrochemical reaction to occur at a measurable rate and its theoretical equilibrium (reversible) potential. It represents the extra energy required to overcome activation barriers.

Question 2082 Baishakh Q2

Problem: An electrochemical cell consists of a nickel electrode dipped in 0.1 M NiSO_4 solution and a silver electrode dipped in 0.05 M AgNO_3 solution. Write the cell notation, cell reaction and calculate the emf of the cell at 30°C . The standard electrode potentials of nickel and silver electrodes are -0.25 V and $+0.80\text{ V}$ respectively.

Solution: Given standard reduction potentials: $E^\circ_{\text{Ni}^{2+}/\text{Ni}} = -0.25\text{ V}$ $E^\circ_{\text{Ag}^+/\text{Ag}} = +0.80\text{ V}$

Since $E^\circ_{\text{Ag}^+/\text{Ag}} > E^\circ_{\text{Ni}^{2+}/\text{Ni}}$, Ag acts as the cathode (reduction) and Ni acts as the anode (oxidation).

Cell Notation: $\text{Ni}(\text{s}) \mid \text{Ni}^{2+}(0.1\text{ M}) \parallel \text{Ag}^+(0.05\text{ M}) \mid \text{Ag}(\text{s})$

Cell Reaction: Anode: $\text{Ni}(\text{s}) \longrightarrow \text{Ni}^{2+} + 2\text{e}^-$ Cathode: $2\text{Ag}^+ + 2\text{e}^- \longrightarrow 2\text{Ag}(\text{s})$ Overall: $\text{Ni}(\text{s}) + 2\text{Ag}^+(\text{aq}) \longrightarrow \text{Ni}^{2+}(\text{aq}) + 2\text{Ag}(\text{s})$ (Here, $n = 2$)

EMF Calculation at 30°C (303 K): Standard Cell Potential:

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$
$$E^\circ_{\text{cell}} = 0.80 - (-0.25) = 1.05\text{ V}$$

Using the Nernst equation for 30°C:

$$\begin{aligned}E_{\text{cell}} &= E_{\text{cell}}^{\circ} - \frac{2.303RT}{nF} \log Q \\ \frac{2.303RT}{F} &= \frac{2.303 \times 8.314 \times 303}{96500} \approx 0.0601 \text{ V} \\ E_{\text{cell}} &= 1.05 - \frac{0.0601}{2} \log \frac{[\text{Ni}^{2+}]}{[\text{Ag}^+]^2} \\ E_{\text{cell}} &= 1.05 - 0.03005 \log \frac{0.1}{(0.05)^2} \\ E_{\text{cell}} &= 1.05 - 0.03005 \log \frac{0.1}{0.0025} \\ E_{\text{cell}} &= 1.05 - 0.03005 \log(40) \\ E_{\text{cell}} &= 1.05 - 0.03005 \times 1.602 \\ E_{\text{cell}} &= 1.05 - 0.0481 = 1.0019 \text{ V}\end{aligned}$$

$$E_{\text{cell}} = 1.002 \text{ V}$$

Question 2082 Baishakh Q2 OR

Problem: Give the working principle of lead acid battery. What are its uses? Also mention its advantages and disadvantages.

Solution: Working Principle: The lead-acid battery operates on the principle of reversible electrochemical reactions. During discharging, it acts as a galvanic cell where the lead (Pb) anode is oxidized to PbSO_4 and the lead dioxide (PbO_2) cathode is reduced to PbSO_4 , consuming sulfuric acid. During charging, an external electrical source forces the reverse reaction (acting as an electrolytic cell), regenerating Pb, PbO_2 , and H_2SO_4 .

Uses: Automobiles (starting, lighting, and ignition), Uninterruptible Power Supplies (UPS), backup power systems, and submarines.

Advantages:

- Capable of delivering high surge currents (useful for engine starters).
- Low cost and reliable technology.
- Highly recyclable.

Disadvantages:

- Low energy-to-weight ratio (heavy).
- Contains toxic lead and corrosive sulfuric acid.
- Sulfation can occur if left in a discharged state for long periods.

Question 2081 Ashwin Q1

Problem: The standard reduction potential of Cd^{2+}/Cd and Ni^{2+}/Ni electrodes are -0.403 V and -0.25 V respectively. Construct a galvanic cell using these electrodes so the standard emf of the cell becomes positive. For what concentration of Cd the emf

of the cell at 25°C becomes zero, if the concentration of Ni is 0.01M.

Solution: Given standard reduction potentials: $E_{\text{Cd}^{2+}/\text{Cd}}^{\circ} = -0.403 \text{ V}$ $E_{\text{Ni}^{2+}/\text{Ni}}^{\circ} = -0.25 \text{ V}$

Cell Construction: For a positive E_{cell}° , the electrode with the higher reduction potential must be the cathode. Cathode: Nickel ($E^{\circ} = -0.25 \text{ V}$) Anode: Cadmium ($E^{\circ} = -0.403 \text{ V}$)
Cell notation: $\text{Cd(s)} \mid \text{Cd}^{2+}(\text{aq}) \parallel \text{Ni}^{2+}(\text{aq}) \mid \text{Ni(s)}$

Standard EMF:

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$$

$$E_{\text{cell}}^{\circ} = -0.25 - (-0.403) = 0.153 \text{ V}$$

Concentration of Cd^{2+} when $E_{\text{cell}} = 0 \text{ V}$: Cell Reaction: $\text{Cd} + \text{Ni}^{2+} \longrightarrow \text{Cd}^{2+} + \text{Ni}$, ($n = 2$) Using Nernst Equation:

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{2} \log \frac{[\text{Cd}^{2+}]}{[\text{Ni}^{2+}]}$$

$$0 = 0.153 - 0.02955 \log \frac{[\text{Cd}^{2+}]}{0.01}$$

$$0.02955 \log \frac{[\text{Cd}^{2+}]}{0.01} = 0.153$$

$$\log \frac{[\text{Cd}^{2+}]}{0.01} = \frac{0.153}{0.02955} = 5.1777$$

$$\frac{[\text{Cd}^{2+}]}{0.01} = 10^{5.1777} = 150543.8$$

$$[\text{Cd}^{2+}] = 150543.8 \times 0.01 = 1505.4 \text{ M}$$

(Note: Such a high concentration is physically impossible, meaning the cell will not reach zero EMF under normal liquid solution conditions just by changing Cd^{2+} concentration, but the mathematical answer is derived above.) $[\text{Cd}^{2+}] = 1505.4 \text{ M}$

Question 2081 Ashwin Q1 OR

Problem: What is Lithium ion battery? Write the reactions involved during discharging and charging of this battery. Write its advantages and disadvantages.

Solution: Lithium-Ion Battery: It is a type of rechargeable battery in which lithium ions move from the negative electrode (anode) to the positive electrode (cathode) during discharge, and back when charging. It typically uses an intercalated lithium compound as the electrode material.

Reactions (with Graphite Anode and LiCoO_2 Cathode): **Discharging:** Anode (Oxidation): $\text{Li}_x \text{C}_6 \longrightarrow x \text{Li}^+ + x \text{e}^- + 6 \text{C}$ Cathode (Reduction): $\text{Li}_{1-x} \text{CoO}_2 + x \text{Li}^+ + x \text{e}^- \longrightarrow \text{LiCoO}_2$ Overall: $\text{Li}_x \text{C}_6 + \text{Li}_{1-x} \text{CoO}_2 \longrightarrow 6 \text{C} + \text{LiCoO}_2$

Charging: The reverse reactions occur, driven by an external power source. Overall: $6 \text{C} + \text{LiCoO}_2 \longrightarrow \text{Li}_x \text{C}_6 + \text{Li}_{1-x} \text{CoO}_2$

Advantages:

- High energy density and lightweight.
- Low self-discharge rate and no memory effect.
- High cell voltage (approx. 3.7 V).

Disadvantages:

- Expensive to manufacture.
- Susceptible to thermal runaway if damaged or overcharged, requiring internal protection circuits.
- Capacity degrades over time (aging).

Chapter 2

Catalysis

Question 2083 Baishakh Q3

Problem: Define homogeneous and heterogeneous catalysis with examples. Discuss the selection criteria of a catalyst.

Solution: Homogeneous Catalysis: The catalytic process in which the catalyst and the reactants are in the same physical phase (usually liquid or gas). *Example:* Oxidation of SO_2 to SO_3 with $\text{NO}(\text{g})$ as a catalyst. $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \xrightarrow{\text{NO}(\text{g})} 2\text{SO}_3(\text{g})$

Heterogeneous Catalysis: The catalytic process in which the catalyst is in a different physical phase from the reactants (usually the catalyst is solid, and reactants are gases or liquids). *Example:* Haber's process for ammonia synthesis using solid Iron catalyst. $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \xrightarrow{\text{Fe}(\text{s})} 2\text{NH}_3(\text{g})$

Selection Criteria of a Catalyst:

- **Activity:** A good catalyst must highly accelerate the rate of a specific chemical reaction.
- **Selectivity:** It must be able to direct a reaction to yield a particular product, avoiding unwanted side products.
- **Stability / Lifespan:** It should be resistant to deactivation by heat, poisoning, or structural changes under operating conditions.
- **Cost and Availability:** It should be economically viable and readily available for industrial scale up.

Question 2082 Bhadra Q3

Problem: Define photocatalysis and electrocatalysis. Explain the photocatalytic process of water splitting.

Solution: Photocatalysis: The acceleration of a photoreaction in the presence of a catalyst. In catalyzed photoreactions, light is absorbed by an adsorbed substrate. In photo-generated catalysis, the photocatalytic activity depends on the ability of the catalyst (usually a semiconductor) to create electron-hole pairs which generate free radicals.

Electrocatalysis: The acceleration of electrochemical reactions at electrode surfaces by using specific electrode materials (electrocatalysts) that lower the activation energy (and thus the overpotential) of the reaction.

Photocatalytic Water Splitting: This is the process of converting water into hydrogen and oxygen using light energy and a photocatalyst (like TiO_2). 1. **Light Absorption:** The semiconductor photocatalyst absorbs photons with energy greater than its bandgap, generating an electron (e^-) in the conduction band and a hole (h^+) in the valence band. $\text{TiO}_2 + h\nu \rightarrow e_{\text{CB}}^- + h_{\text{VB}}^+$ 2. **Oxidation of Water:** The photogenerated holes migrate to the surface and oxidize water molecules to produce oxygen gas. $2\text{H}_2\text{O} + 4h^+ \longrightarrow \text{O}_2 + 4\text{H}^+$ 3. **Reduction of Water:** The photogenerated electrons reduce the hydrogen ions to produce hydrogen gas. $4\text{H}^+ + 4e^- \longrightarrow 2\text{H}_2$ Overall Reaction: $2\text{H}_2\text{O} \xrightarrow{h\nu, \text{catalyst}} 2\text{H}_2 + \text{O}_2$

Question 2082 Baishakh Q3

Problem: Discuss any four selection criteria of a catalyst. Explain the phenomenon of electrocatalytic water splitting.

Solution: Selection Criteria of a Catalyst: 1. **High Activity:** It must significantly increase the reaction rate. 2. **High Selectivity:** It must direct the reaction to the desired product to maximize yield and minimize waste. 3. **Poison Resistance:** It must not easily deactivate in the presence of trace impurities (poisons). 4. **Regenerability and Cost:** It should be easy to regenerate once deactivated, and the overall cost must be viable for industry.

Electrocatalytic Water Splitting: This is the decomposition of water into hydrogen and oxygen by passing an electric current, utilizing electrocatalysts to lower the overpotential required for the reactions. The process consists of two half-reactions:

- **Hydrogen Evolution Reaction (HER)** at the cathode: $2\text{H}^+ + 2e^- \longrightarrow \text{H}_2$ (Electrocatalysts like Pt are commonly used to lower the energy barrier.)
- **Oxygen Evolution Reaction (OER)** at the anode: $2\text{H}_2\text{O} \longrightarrow \text{O}_2 + 4\text{H}^+ + 4e^-$ (Electrocatalysts like IrO_2 or RuO_2 are used here as this reaction has sluggish kinetics and high overpotential.)

The catalysts improve efficiency by providing alternative pathways with lower activation energies, thus saving electrical energy.

Question 2081 Ashwin Q3

Problem: How are the catalysts used in the field of pollution control? Explain. Write any two examples of biocatalyst used in degradation of dyes.

Solution: Catalysts in Pollution Control: Catalysts play a crucial role in converting harmful pollutants into benign substances. The most prominent example is the **Catalytic Converter** in automobile exhausts. It contains catalysts like Platinum (Pt), Palladium (Pd), and Rhodium (Rh).

- It reduces nitrogen oxides (NO_x) to elemental nitrogen and oxygen. ($2\text{NO}_x \longrightarrow x\text{O}_2 + \text{N}_2$)

- It oxidizes carbon monoxide (CO) to carbon dioxide (CO₂). ($2\text{CO} + \text{O}_2 \longrightarrow 2\text{CO}_2$)
- It oxidizes unburnt hydrocarbons to CO₂ and water. ($\text{C}_x\text{H}_y + (x + y/4)\text{O}_2 \longrightarrow x\text{CO}_2 + y/2\text{H}_2\text{O}$)

Catalysts are also used in industry for removing SO₂ and NO_x from flue gases (e.g., Selective Catalytic Reduction - SCR, which uses ammonia over a catalyst to convert NO_x to N₂ and water).

Biocatalysts used in degradation of dyes: Biocatalysts (enzymes) are employed in bioremediation of synthetic dyes in wastewater. Two examples are: 1. **Laccases:** They oxidize phenolic and non-phenolic dyes by a radical mechanism. 2. **Peroxidases (e.g., Lignin Peroxidase or Horseradish Peroxidase):** They use hydrogen peroxide to catalyze the oxidation and degradation of complex dye molecules.

Chapter 3

Analytical Chemistry

Question 2083 Baishakh Q2

Problem: Explain, how the acidic buffer solution resists the change in pH on addition of small amount of acid or base. What is the pH of resulting mixture obtained by mixing 100 ml of 0.2 M HCl and 50 ml of 0.5 M NH_4OH solution. (K_b for ammonia = 1.8×10^{-5}).

Solution: Mechanism of Acidic Buffer: An acidic buffer consists of a weak acid (e.g., CH_3COOH) and its salt with a strong base (e.g., CH_3COONa). In solution: $\text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}^+$ (slightly ionized) $\text{CH}_3\text{COONa} \longrightarrow \text{CH}_3\text{COO}^- + \text{Na}^+$ (completely ionized)

- **Addition of acid (H^+):** The added H^+ ions combine with the large reserve of CH_3COO^- ions (from the salt) to form weakly ionized acetic acid. $\text{CH}_3\text{COO}^- + \text{H}^+ \longrightarrow \text{CH}_3\text{COOH}$ Thus, free H^+ does not accumulate, and pH remains nearly constant.
- **Addition of base (OH^-):** The added OH^- ions react with the free H^+ in the solution to form water. More CH_3COOH ionizes to replace the consumed H^+ . $\text{CH}_3\text{COOH} + \text{OH}^- \longrightarrow \text{CH}_3\text{COO}^- + \text{H}_2\text{O}$ Thus, pH remains nearly constant.

Numerical Calculation: Initial moles of HCl = $100 \text{ ml} \times 0.2 \text{ M} = 20 \text{ mmol}$ Initial moles of $\text{NH}_4\text{OH} = 50 \text{ ml} \times 0.5 \text{ M} = 25 \text{ mmol}$ Reaction: $\text{NH}_4\text{OH} + \text{HCl} \longrightarrow \text{NH}_4\text{Cl} + \text{H}_2\text{O}$ Since NH_4OH is in excess: Moles of NH_4OH left = $25 - 20 = 5 \text{ mmol}$ Moles of NH_4Cl formed = 20 mmol Total volume = $100 + 50 = 150 \text{ ml}$

The resulting mixture is a basic buffer ($\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$). $pK_b = -\log(1.8 \times 10^{-5}) = 4.74$ Using Henderson-Hasselbalch equation for basic buffers:

$$\begin{aligned} pOH &= pK_b + \log \left(\frac{[\text{Salt}]}{[\text{Base}]} \right) \\ pOH &= 4.74 + \log \left(\frac{20/150}{5/150} \right) \\ pOH &= 4.74 + \log(4) \\ pOH &= 4.74 + 0.602 = 5.342 \end{aligned}$$

$$\begin{aligned} pH &= 14 - pOH \\ pH &= 14 - 5.342 = 8.658 \end{aligned}$$

Question 2083 Baishakh Q4

Problem: Define chromatographic technique. Write the principle and application of thin layer chromatography.

Solution: Chromatographic Technique: It is an analytical technique used for the separation, identification, and purification of components of a mixture based on their differential distribution between two phases: a stationary phase and a mobile phase.

Principle of Thin Layer Chromatography (TLC): TLC is a solid-liquid technique based primarily on the principle of *adsorption*. The stationary phase is a thin layer of an adsorbent (like silica gel or alumina) coated on a flat inert plate. The mobile phase is a liquid solvent that moves up the plate by capillary action. Components of the mixture move at different rates depending on their relative affinities for the stationary phase (adsorption) and the mobile phase (solubility).

Applications of TLC:

- Checking the purity of a given compound.
 - Identifying compounds by comparing their retention factor (R_f) values with standard references.
 - Monitoring the progress of a chemical reaction.
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Question 2083 Baishakh Q4 OR

Problem: Write down the principle of IR spectroscopy. Discuss the significance of functional group region and fingerprint region in IR spectrum.

Solution: Principle of IR Spectroscopy: IR spectroscopy is based on the absorption of infrared radiation by molecules, which causes transitions between their vibrational energy levels. Only those molecules that exhibit a net change in dipole moment during vibration are "IR active." Different chemical bonds have distinct vibrational frequencies, meaning they absorb characteristic wavelengths of IR light.

Functional Group Region (4000 cm^{-1} to 1500 cm^{-1}): In this region, the absorption bands are typically due to stretching vibrations of specific functional groups (e.g., -OH, -C=O, -N-H, -C≡C-). It is highly significant for identifying the presence or absence of specific functional groups in an unknown molecule.

Fingerprint Region (1500 cm^{-1} to 400 cm^{-1}): This region contains complex, highly characteristic bending vibrations and skeletal vibrations of the molecule as a whole. Even if two different molecules have the same functional groups, their fingerprint regions will be unique. Thus, it is significant for the exact identification of a compound by comparing it with the spectrum of a known standard.

Question 2082 Bhadra Q2

Problem: Derive the Henderson's equation for basic buffer. If 3.0 milli mole of NaOH is added to 100 ml of a buffer which is 0.10 M in acetic acid and 0.1 M in sodium acetate, what will be the change in pH of solution? [$pK_a = 4.74$].

Solution: Derivation of Henderson-Hasselbalch Equation for Basic Buffer: Consider a basic buffer consisting of a weak base BOH and its salt BA . The weak base dissociates partially: $BOH \rightleftharpoons B^+ + OH^-$ The dissociation constant K_b is:

$$K_b = \frac{[B^+][OH^-]}{[BOH]}$$

Rearranging for $[OH^-]$:

$$[OH^-] = K_b \frac{[BOH]}{[B^+]}$$

Since the salt BA dissociates completely ($BA \longrightarrow B^+ + A^-$), the concentration of B^+ is essentially equal to the salt concentration. Also, $[BOH]$ is approximately the initial concentration of the base.

$$[OH^-] = K_b \frac{[Base]}{[Salt]}$$

Taking negative logarithm on both sides:

$$-\log[OH^-] = -\log K_b - \log \left(\frac{[Base]}{[Salt]} \right)$$

$$pOH = pK_b + \log \left(\frac{[Salt]}{[Base]} \right)$$

Numerical Calculation: Initial Buffer (Acidic): Volume = 100 ml $[Acid] = 0.10 \text{ M} \Rightarrow$ Moles of Acid = 10 mmol $[Salt] = 0.10 \text{ M} \Rightarrow$ Moles of Salt = 10 mmol Initial pH: $pH_1 = pK_a + \log \left(\frac{[Salt]}{[Acid]} \right) = 4.74 + \log \left(\frac{10}{10} \right) = 4.74$

After adding 3.0 mmol of NaOH: NaOH reacts with the weak acid ($CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$). Moles of Acid remaining = $10 - 3 = 7$ mmol Moles of Salt formed = $10 + 3 = 13$ mmol New pH:

$$pH_2 = 4.74 + \log \left(\frac{13}{7} \right)$$

$$pH_2 = 4.74 + 0.269 = 5.009$$

Change in pH:

$$\Delta pH = pH_2 - pH_1 = 5.009 - 4.74 = 0.269$$

$\Delta pH = +0.269$

Question 2082 Bhadra Q4

Problem: Write the principle of chromatography along with its applications. How is X-ray diffraction method useful in the analysis of crystalline structure?

Solution: Principle of Chromatography: (As defined above, it relies on the differential distribution/partitioning of components between a stationary phase and a mobile phase based on adsorption or solubility).

Applications: - Separation of complex mixtures (e.g., amino acids, plant pigments). - Purification of pharmaceutical products. - Forensic testing (detecting drugs in bodily fluids).

X-Ray Diffraction (XRD) in Crystalline Analysis: XRD is based on Bragg's Law ($n\lambda = 2d \sin \theta$). When monochromatic X-rays strike a crystal, the regular arrangement of atoms acts as a 3D diffraction grating. The angles and intensities of the diffracted beams are measured. It is useful because: - It allows determination of the exact crystal lattice structure (unit cell dimensions). - It can identify the arrangement of atoms and bond lengths in crystalline materials. - It differentiates between various polymorphic forms of a substance.

Question 2082 Bhadra Q4 OR

Problem: Describe the principle of NMR spectroscopy. Draw and explain NMR spectrum of ethanol.

Solution: Principle of NMR Spectroscopy: Nuclear Magnetic Resonance (NMR) spectroscopy is based on the magnetic properties of certain atomic nuclei (like ^1H or ^{13}C). When placed in a strong external magnetic field, these nuclei align either with or against the field. Absorption of radio-frequency (RF) radiation causes the nuclei to "flip" their spins (resonance). The exact frequency absorbed depends on the electron cloud surrounding the nucleus (chemical shift), providing details about the chemical environment of the atoms.

NMR Spectrum of Ethanol ($\text{CH}_3-\text{CH}_2-\text{OH}$): (Draw a spectrum with three distinct signal clusters). The ^1H -NMR spectrum of ethanol shows three sets of peaks due to three different types of protons:

1. **-OH proton:** A singlet (typically broad) around $\delta = 2.0 - 5.0$ ppm. It does not usually split because of rapid chemical exchange.
2. **-CH₂- protons:** A quartet around $\delta = 3.6$ ppm. It is split into four peaks ($n + 1$ rule) by the adjacent 3 protons of the methyl group.
3. **-CH₃ protons:** A triplet around $\delta = 1.2$ ppm. It is split into three peaks by the adjacent 2 protons of the methylene group.

Question 2082 Baishakh Q3a

Problem: How does a solution containing the mixture of benzoic acid and sodium benzoate maintain its constant pH value even on the addition of small amount of strong acid or alkali? Explain.

Solution: A mixture of benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$, weak acid) and sodium benzoate ($\text{C}_6\text{H}_5\text{COONa}$, its conjugate base salt) forms an acidic buffer.

- **When strong acid (H^+) is added:** The added H^+ is neutralized by the benzoate ion ($\text{C}_6\text{H}_5\text{COO}^-$) from the salt to form undissociated benzoic acid. $\text{C}_6\text{H}_5\text{COO}^- + \text{H}^+ \longrightarrow \text{C}_6\text{H}_5\text{COOH}$

- **When strong alkali (OH^-) is added:** The added OH^- is neutralized by the benzoic acid to form water and benzoate ion. $\text{C}_6\text{H}_5\text{COOH} + \text{OH}^- \longrightarrow \text{C}_6\text{H}_5\text{COO}^- + \text{H}_2\text{O}$

Because the added H^+ or OH^- are consumed and converted into components already present in the buffer, the ratio of $[\text{Salt}]/[\text{Acid}]$ changes only slightly, keeping the pH nearly constant.

Question 2082 Baishakh Q3b

Problem: Calculate the concentration of sodium formate, that must be present in 0.10 M solution of formic acid to prepare a buffer solution of pH 3.80. K_a for formic acid is 1.8×10^{-4} .

Solution: Given: $\text{pH} = 3.80$ $[\text{Acid (Formic acid)}] = 0.10 \text{ M}$ $K_a = 1.8 \times 10^{-4} \Rightarrow \text{p}K_a = -\log(1.8 \times 10^{-4}) = 3.74$ Using Henderson-Hasselbalch equation:

$$\begin{aligned}\text{pH} &= \text{p}K_a + \log \left(\frac{[\text{Salt}]}{[\text{Acid}]}\right) \\ 3.80 &= 3.74 + \log \left(\frac{[\text{Salt}]}{0.10}\right) \\ 0.06 &= \log \left(\frac{[\text{Salt}]}{0.10}\right) \\ 10^{0.06} &= \frac{[\text{Salt}]}{0.10} \\ 1.148 &= \frac{[\text{Salt}]}{0.10} \\ [\text{Salt}] &= 0.1148 \text{ M}\end{aligned}$$

$[\text{Sodium formate}] = 0.115 \text{ M}$

Question 2082 Baishakh Q4

Problem: Write the principle and applications of NMR spectroscopy. Why is tetramethylsilane used as a reference compound in NMR spectroscopy?

Solution: Principle and Applications: (Refer to 2082 Bhadra Q4 OR for principle). Applications include structural elucidation of organic compounds, determining the purity of samples, and medical imaging (MRI).

Why Tetramethylsilane (TMS) is used as a reference:

- It has 12 equivalent protons, producing a single, intense, sharp peak.
- Its protons are highly shielded due to the low electronegativity of Silicon, so its signal appears at a lower frequency than almost all other organic protons. By convention, it is assigned $\delta = 0$ ppm.
- It is chemically inert, non-toxic, and highly volatile (easy to remove from the sample).

Question 2082 Baishakh Q4 OR

Problem: a) Explain the principle involved in the separation of components of a mixture by paper chromatography. b) Show your acquaintance to hypsochromic shift and bathochromic shift in UV visible spectroscopy.

Solution: a) **Principle of Paper Chromatography:** It is a type of partition chromatography. The stationary phase is the water trapped within the cellulose fibers of the filter paper, while the mobile phase is an organic solvent that travels up the paper by capillary action. Components of the mixture are separated based on their differential partition coefficients between the stationary aqueous phase and the mobile organic phase.

b) **Shifts in UV-Visible Spectroscopy:**

- **Bathochromic Shift (Red Shift):** A shift of the absorption maximum ($\lambda_{\max}$) to a longer wavelength (lower energy). This is often caused by the presence of an auxochrome, conjugation, or a change in solvent polarity.
- **Hypsochromic Shift (Blue Shift):** A shift of the absorption maximum ($\lambda_{\max}$) to a shorter wavelength (higher energy). It is typically caused by the removal of conjugation or a change in solvent polarity.

Question 2081 Ashwin Q2

Problem: Taking suitable example, explain how the basic buffer solution resists the change in pH with the addition of small amount of acid or base.

Solution: (See 2083 Baishakh Q2 for basic buffer mechanism conceptually, replacing acidic with basic context). A basic buffer consists of a weak base (e.g., NH_4OH) and its salt (e.g., NH_4Cl).

- Add acid (H^+): $\text{NH}_4\text{OH} + \text{H}^+ \longrightarrow \text{NH}_4^+ + \text{H}_2\text{O}$
- Add base (OH^-): $\text{NH}_4^+ + \text{OH}^- \longrightarrow \text{NH}_4\text{OH}$

Both added ions are neutralized, maintaining pH.

Question 2081 Ashwin Q4

Problem: What is Chromatography? Explain the principle and application of thin layer chromatography. How does it vary from paper chromatography?

Solution: Chromatography is a separation technique. (See previous answers for TLC principle and applications). **Difference from Paper Chromatography:** TLC uses a solid adsorbent (like silica gel) spread on a rigid plate as the stationary phase (acting via adsorption), whereas paper chromatography uses water trapped in cellulose fibers as the stationary phase (acting via partition). TLC is generally faster, provides better resolution, and handles a wider variety of solvents and corrosive visualizing agents than paper chromatography.

Chapter 4

Complexes and Alloys

Question 2083 Baishakh Q5

Problem: What is chelate? Using VBT, predict the geometry and magnetic property of $[\text{CoF}_6]^{3-}$. Write the formula of following co-ordination compounds: i) Triammine-triaquachromium (III) chloride ii) Potassium trioxalatoaluminate (III)

Solution: **Chelate:** A chelate is a cyclic complex formed when a multidentate ligand coordinates to a central metal atom through two or more donor atoms simultaneously.

VBT on $[\text{CoF}_6]^{3-}$: Central metal: Co (atomic number 27). Electronic configuration: $[\text{Ar}] 3d^7 4s^2$. In $[\text{CoF}_6]^{3-}$, Oxidation state of Co is +3. Co^{3+} : $[\text{Ar}] 3d^6$. Ligand: F^- is a weak field ligand, so it does not force pairing of electrons in the 3d orbitals. Since 3d orbitals remain unpaired, the complex uses outer 4d orbitals for hybridization. Hybridization: sp^3d^2 (Outer orbital complex) Geometry: **Octahedral** Magnetic Property: Contains 4 unpaired electrons in the 3d subshell. Hence, it is **highly paramagnetic**.

Formulas: i) Triamminetriaquachromium (III) chloride: $[\text{Cr}(\text{NH}_3)_3(\text{H}_2\text{O})_3]\text{Cl}_3$ ii) Potassium trioxalatoaluminate (III): $\text{K}_3[\text{Al}(\text{C}_2\text{O}_4)_3]$

Question 2083 Baishakh Q6

Problem: a) What are the postulates of crystal field theory? Explain crystal field splitting of d-orbitals in an octahedral complex. b) Give an applications of aluminum alloy.

Solution: **Postulates of Crystal Field Theory (CFT):**

- The central metal ion and ligands are treated as point charges (or point dipoles).
- The interaction between the metal ion and the ligands is purely electrostatic.
- The five d-orbitals of the isolated gaseous metal ion are degenerate (have equal energy).
- When ligands approach the central metal ion, their negative fields repel the electrons in the metal's d-orbitals, breaking their degeneracy and causing them to split into different energy levels.

Splitting in Octahedral Complex: In an octahedral field, six ligands approach the central metal along the x, y, and z axes. The $d_{x^2-y^2}$ and d_{z^2} orbitals point directly along these axes,

experiencing greater repulsion from the ligands, thus their energy is raised (forming the e_g set). The d_{xy} , d_{yz} , and d_{zx} orbitals point between the axes, experiencing less repulsion, so their energy is lowered (forming the t_{2g} set). The energy difference between the e_g and t_{2g} sets is called the crystal field splitting energy, Δ_o .

Applications of Aluminum Alloy: Used in aerospace engineering (aircraft body parts) due to its high strength-to-weight ratio and corrosion resistance (e.g., Duralumin).

Question 2082 Bhadra Q5

Problem: On the basis of VBT, explain the structure and hybridization involved in $[\text{Ni}(\text{CO})_4]$. Write the IUPAC names of followings complexes: i) $[\text{Cr}(\text{en})_3]\text{Br}_3$ ii) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2$ iii) $\text{K}_3[\text{AlF}_6]$ iv) $[\text{Cr}(\text{NH}_3)_4\text{SO}_4]\text{NO}_3$

Solution: VBT on $[\text{Ni}(\text{CO})_4]$: Oxidation state of Ni is 0. Electronic configuration of Ni: $[\text{Ar}] 3d^8 4s^2$ CO is a very strong field ligand. It forces the pairing of 4s electrons into the 3d orbitals. After pairing, the 3d subshell is completely filled ($3d^{10}$). The available empty orbitals for hybridization are one 4s and three 4p orbitals. Hybridization: sp^3 Geometry: **Tetrahedral** Magnetic Property: Diamagnetic (no unpaired electrons).

IUPAC Names: i) $[\text{Cr}(\text{en})_3]\text{Br}_3$: Tris(ethylenediamine)chromium(III) bromide ii) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2$: Pentaquachloridochromium(III) chloride iii) $\text{K}_3[\text{AlF}_6]$: Potassium hexafluoroaluminate(III) iv) $[\text{Cr}(\text{NH}_3)_4\text{SO}_4]\text{NO}_3$: Tetraamminesulfatochromium(III) nitrate

Question 2082 Bhadra Q6

Problem: Explain crystal field splitting of d orbitals in tetrahedral complex. Also describe the color formation process in complexes with an example.

Solution: Splitting in Tetrahedral Complex: In a tetrahedral complex, four ligands approach the central metal ion between the axes. Consequently, the d_{xy} , d_{yz} , and d_{zx} orbitals (which point between the axes) experience more repulsion than the $d_{x^2-y^2}$ and d_{z^2} orbitals (which point along the axes). This reverses the splitting pattern of an octahedral field: the t_2 set is raised in energy, and the e set is lowered in energy. The splitting energy (Δ_t) is approximately $4/9$ of Δ_o .

Color Formation Process: According to CFT, color in transition metal complexes arises from **d-d transitions**. When white light falls on a complex, an electron from a lower energy d-orbital (e.g., t_{2g}) absorbs a specific wavelength of light to jump to a higher energy d-orbital (e.g., e_g). The transmitted or reflected light corresponds to the complementary color of the absorbed light. *Example:* In $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$, the Ti^{3+} ion has one d-electron in the t_{2g} level. It absorbs green-yellow light from the visible spectrum to jump to the e_g level. The complex appears purple (the complementary color).

Question 2082 Baishakh Q5 & Q6

Problem: a) $[\text{NiCl}_4]^{2-}$ is tetrahedral and paramagnetic while $[\text{Ni}(\text{CN})_4]^{2-}$ is square planar and diamagnetic, Explain with reference to VBT. Write the IUPAC name: i) $\text{Na}_3[\text{AgF}_4]$ ii) $[\text{Co}(\text{en})_3]\text{Br}_3$. b) Define alloys with examples. Give application of

aluminum alloys.

Solution: $[\text{NiCl}_4]^{2-}$: Ni is in +2 state ($3d^8$). Cl^- is a weak field ligand, so it does not pair the d electrons. Two unpaired electrons remain in 3d. Hybridization is sp^3 , making it tetrahedral and paramagnetic. $[\text{Ni}(\text{CN})_4]^{2-}$: Ni is in +2 state ($3d^8$). CN^- is a strong field ligand, forcing the two unpaired electrons in 3d to pair up. This leaves one 3d orbital empty. Hybridization is dsp^2 , making it square planar and diamagnetic.

IUPAC Names: i) $\text{Na}_3[\text{AgF}_4]$: Sodium tetrafluoroargentate(I) (*Note: Ag(I) typically forms linear, but this compound name follows IUPAC rules based on the formula provided*)
ii) $[\text{Co}(\text{en})_3]\text{Br}_3$: Tris(ethylenediamine)cobalt(III) bromide

Alloys: An alloy is a homogeneous mixture (solid solution) of two or more metals, or a metal and a non-metal, designed to improve the physical properties of the base metal. *Examples:* Steel (Fe + C), Brass (Cu + Zn). *Applications of Aluminum alloys:* Used in the construction of aircraft frames and automobile parts due to high strength and low density.

Question 2081 Ashwin Q5

Problem: a) What are inner and outer orbital octahedral complexes? On the basis of VBT predict the magnetism and geometry of $[\text{Fe}(\text{CN})_6]^{4-}$. b) From the given IUPAC names write the molecular formula: i) Potassiumtetraiodomercurate(II) ii) Dichloro-bis-ethylenediaminenickel (IV) sulphate iii) Tetraaquadichlorocobalt (III) chloride

Solution: **Inner vs. Outer Orbital Complexes:** - **Inner Orbital Complex:** The complex uses the inner $(n-1)d$ orbitals for hybridization (e.g., d^2sp^3). Usually formed by strong field ligands. - **Outer Orbital Complex:** The complex uses the outer nd orbitals for hybridization (e.g., sp^3d^2). Usually formed by weak field ligands.

$[\text{Fe}(\text{CN})_6]^{4-}$: Fe is in +2 oxidation state: $[\text{Ar}] 3d^6$. CN^- is a strong ligand, causing the six electrons in 3d to pair completely, leaving two 3d orbitals empty. Hybridization is d^2sp^3 . Geometry: **Octahedral**. Magnetism: **Diamagnetic** (0 unpaired electrons).

Molecular Formulas: i) Potassium tetraiodomercurate(II): $\text{K}_2[\text{HgI}_4]$ ii) Dichlorido-bis(ethylenediamine)nickel(IV) sulphate: $[\text{NiCl}_2(\text{en})_2](\text{SO}_4)$ iii) Tetraaquadichloridocobalt(III) chloride: $[\text{CoCl}_2(\text{H}_2\text{O})_4]\text{Cl}$

Question 2081 Ashwin Q6

Problem: What are metallic alloys? Give examples with their uses. Using CFT, explain the color formation and magnetic property of $[\text{Cu}(\text{NH}_3)_4]^{2+}$.

Solution: **Metallic Alloys:** Solid solutions of metals that provide enhanced properties over pure metals (like increased hardness, corrosion resistance). Example: Bronze (Cu + Sn) used for statues and medals; Stainless Steel (Fe + Cr + Ni) used for cutlery.

$[\text{Cu}(\text{NH}_3)_4]^{2+}$ **according to CFT:** Cu^{2+} has a $3d^9$ configuration. In a square planar field (due to NH_3), the d-orbitals split heavily, with the highest energy orbital ($d_{x^2-y^2}$) containing one unpaired electron. - **Color:** The single unpaired electron can undergo d-d transition to a higher energy level by absorbing red/orange light from the visible spectrum, transmitting a deep blue color. - **Magnetic Property:** Since there is exactly one unpaired electron, the

complex is **paramagnetic**.

Chapter 5

Sustainable Water and Environment

Question 2083 Baishakh Q7a

Problem: What is BOD and COD and their consequences with reference to water pollution? Discuss the steps involved in the industrial treatment of wastewater.

Solution: **BOD (Biochemical Oxygen Demand):** The amount of dissolved oxygen required by aerobic microorganisms to break down organic matter in a given volume of water at a certain temperature over a specific time (usually 5 days at 20°C). **COD (Chemical Oxygen Demand):** The amount of oxygen required to chemically oxidize all organic and inorganic oxidizable substances in water using a strong chemical oxidant (e.g., potassium dichromate). COD is always greater than or equal to BOD.

Consequences in Water Pollution: High BOD/COD indicates a high level of organic pollution. As organic matter decomposes, oxygen is depleted from the water body, suffocating aquatic life (fish kills) and leading to anaerobic conditions which produce foul-smelling gases like H_2S and methane.

Industrial Wastewater Treatment Steps: 1. **Primary Treatment (Physical/Chemical):** Removal of large suspended solids via screening, grit chambers, and primary sedimentation tanks. Coagulation/flocculation may be added to settle fine particles. 2. **Secondary Treatment (Biological):** Aerobic or anaerobic microorganisms consume the dissolved organic matter. Examples include activated sludge processes, trickling filters, and oxidation ponds. 3. **Tertiary Treatment (Advanced):** Removal of specific pollutants like heavy metals, nitrogen, and phosphorus using chemical precipitation, reverse osmosis, or carbon adsorption. Followed by disinfection (chlorination or UV).

Question 2083 Baishakh Q7b

Problem: Describe your idea about the segregation and management of solid waste. Also, discuss the challenges to be faced during the process of waste to energy implementation.

Solution: **Segregation and Management:** Solid waste management relies on the 3R principle (Reduce, Reuse, Recycle). Effective management begins with source segregation

into: - Biodegradable (organic food waste) → Used for composting or biogas. - Recyclables (paper, plastic, glass, metal) → Sent to recycling facilities. - Hazardous (batteries, medical waste, e-waste) → Special incineration or secure landfilling. - Inert/Residual → Sent to engineered sanitary landfills.

Challenges in Waste-to-Energy (WtE): WtE processes (like incineration, gasification, or pyrolysis) convert non-recyclable waste into electricity or heat. Challenges include: 1. **High Moisture Content:** Organic waste often has high water content, lowering its calorific value and making incineration inefficient. 2. **Toxic Emissions:** Burning plastics and electronic components can release dioxins, furans, and heavy metals, requiring expensive emission control systems. 3. **High Capital Cost:** Setting up and maintaining advanced WtE plants is highly expensive compared to traditional landfilling. 4. **Improper Segregation:** Mixed waste clogs and damages machinery.

Question 2082 Bhadra Q7a

Problem: What is green chemistry? Explain two principles of green chemistry with suitable examples. Discuss COD and its consequences with reference to water pollution.

Solution: Green Chemistry: The design of chemical products and processes that reduce or eliminate the use and generation of hazardous substances.

Two Principles: 1. **Prevention of Waste:** It is better to prevent waste than to treat or clean up waste after it is formed. *Example:* Using a synthesis route with a high atom economy so all reactants end up in the final product. 2. **Use of Safer Solvents:** The use of auxiliary substances (e.g., solvents, separation agents) should be made unnecessary wherever possible and innocuous when used. *Example:* Replacing toxic organic solvents (like benzene) with water or supercritical CO₂.

COD and Consequences: (See 2083 Baishakh Q7a above. It indicates chemical pollution burden, leading to oxygen depletion).

Question 2082 Bhadra Q7b

Problem: Discuss the impacts of SO_x and NO_x on the environment and human health? How can we control emission containing SO_x?

Solution: Impacts of SO_x and NO_x:

- **Environment:** They react with water, oxygen, and other chemicals in the atmosphere to form sulfuric and nitric acids, leading to **Acid Rain**. This damages crops, forests, historical monuments (like the Taj Mahal), and acidifies lakes, killing aquatic life. NO_x also contributes to photochemical smog.
- **Human Health:** Inhalation aggravates respiratory diseases like asthma, bronchitis, and decreases lung function.

Control of SO_x Emission: - **Pre-combustion:** Using low-sulfur coal or desulfurizing fossil fuels before burning. - **Flue Gas Desulfurization (FGD):** Scrubbing the exhaust gas with alkaline substances like limestone (CaCO₃) or lime (CaO) to form calcium sulfate (gypsum). $\text{CaCO}_3 + \text{SO}_2 + 0.5 \text{O}_2 \longrightarrow \text{CaSO}_4 + \text{CO}_2$

Question 2082 Baishakh Q7

Problem: a) Explain about any four principles of green chemistry. What are heavy metal ions? Discuss any suitable method for the removal of heavy metal ions from industrial wastewater. b) What are air pollutants? How do particulate matter (PM) and Greenhouse gases (GHGs) cause air pollution? Write their effects and remedies.

Solution: a) **Four Principles of Green Chemistry:** 1. *Waste Prevention:* Better to prevent waste than clean it up. 2. *Atom Economy:* Maximize incorporation of all materials into the final product. 3. *Less Hazardous Chemical Syntheses:* Use and generate substances with little or no toxicity. 4. *Design for Energy Efficiency:* Conduct processes at ambient temperature and pressure.

Heavy Metal Ions & Removal: Heavy metal ions (like Pb^{2+} , Hg^{2+} , Cd^{2+} , Cr^{6+}) are highly toxic, non-biodegradable elements. *Removal Method (Chemical Precipitation):* The pH of wastewater is raised using lime ($\text{Ca}(\text{OH})_2$) or caustic soda, converting soluble metal ions into insoluble metal hydroxides, which precipitate out and are removed by sedimentation.
$$\text{M}^{2+} + 2\text{OH}^- \longrightarrow \text{M}(\text{OH})_2 \downarrow$$

b) **Air Pollutants:** Substances in the air that can cause harm to humans, flora, fauna, and materials. - **Particulate Matter (PM):** Tiny solid/liquid particles suspended in air (PM10, PM2.5). They penetrate deep into lungs, causing respiratory/cardiovascular diseases, and reduce visibility. *Remedy:* Use electrostatic precipitators and bag filters in industries. - **Greenhouse Gases (GHGs):** Gases like CO_2 , CH_4 , N_2O . They trap infrared radiation from the Earth's surface, leading to Global Warming and Climate Change. *Remedy:* Shift to renewable energy, plant trees, and reduce fossil fuel usage.

Question 2081 Ashwin Q7

Problem: What is the goal of green chemistry? Mention any four principles of green chemistry. [Missing text on water pollution]

Solution: **Goal:** To create better, safer chemicals while choosing the safest, most efficient ways to synthesize them and to reduce wastes at the source. **Four Principles:** (Refer to the previous answer: Prevention, Atom Economy, Safer Solvents, Energy Efficiency). (Note: Additional parts regarding pollution can be addressed by applying standard concepts of BOD/COD and wastewater treatment).

Chapter 6

Nanoscience

Question 2083 Baishakh Q8

Problem: Classify nanomaterials based on their dimensions including examples. Explain Chemical Vapor Deposition (CVD) process in detail.

Solution: Classification of Nanomaterials based on Dimensions: Nanomaterials are classified based on the number of dimensions (x, y, z) that are outside the nanoscale range (> 100 nm).

- **0D (Zero-Dimensional):** All three dimensions are in the nanoscale (< 100 nm). Example: Quantum dots, Fullerene (C_{60}), nanoparticles.
- **1D (One-Dimensional):** Two dimensions are in the nanoscale, and one dimension is extended (macroscopic). Example: Carbon nanotubes (CNTs), nanowires, nanofibers.
- **2D (Two-Dimensional):** One dimension is in the nanoscale (thickness), while two dimensions are extended. Example: Graphene, MXenes, thin films.
- **3D (Three-Dimensional):** None of the dimensions are in the nanoscale, but the material possesses a nanoscale internal structure or is composed of nanoblocks. Example: Bulk nanostructured materials, mesoporous silica.

Chemical Vapor Deposition (CVD): CVD is a "bottom-up" chemical process used to produce high-purity, high-performance solid nanomaterials (like carbon nanotubes and graphene) from the gas phase. **Process:** 1. *Precursor delivery:* A volatile precursor gas containing the desired material elements is introduced into a heated reaction chamber. 2. *Reaction:* As the gas passes over or comes into contact with a heated substrate (often coated with a catalyst like Ni or Fe), it decomposes or reacts. 3. *Deposition:* The non-volatile solid reaction product deposits atom-by-atom onto the substrate, forming a thin film or nanostructure. 4. *Exhaust:* The volatile by-products are removed from the chamber by gas flow. **Advantages:** High purity, uniform coating on complex shapes, and scalable.

Question 2082 Bhadra Q8

Problem: Define nanofibers and Mxene. Discuss physical vapor deposition (PVD) method for preparation of nanomaterials.

Solution: Nanofibers: These are 1D nanomaterials with diameters in the nanometer

range (typically less than 100 nm) and lengths much larger. They have an extremely high surface-area-to-volume ratio and high porosity, making them useful in filtration and tissue engineering. They are often synthesized via electrospinning.

MXene: These are a class of 2D inorganic compounds consisting of transition metal carbides, nitrides, or carbonitrides (e.g., Ti_3C_2). They combine the metallic conductivity of transition metals with the hydrophilic nature of their hydroxyl/oxygen terminated surfaces.

Physical Vapor Deposition (PVD): PVD is a "top-down" vacuum deposition method used to produce thin films and coatings. It involves purely physical processes (no chemical reactions). **Mechanism:** 1. *Vaporization:* The bulk source material is converted into a vapor through physical means, either by heating (thermal evaporation) or by bombarding with ions (sputtering). 2. *Transportation:* The vaporized atoms/molecules travel across a vacuum or low-pressure gaseous environment to the substrate. 3. *Condensation:* The vapor condenses on the cooler substrate surface to form a thin nanostructured film.

Question 2082 Baishakh Q8

Problem: How is carbon nanotube different from graphene? Discuss nanomaterials of various dimensions with examples.

Solution: Differences between CNT and Graphene:

- **Structure:** Graphene is a single, flat, 2D sheet of sp^2 -hybridized carbon atoms arranged in a honeycomb lattice. A Carbon Nanotube (CNT) is a 1D cylinder made by rolling up a graphene sheet.
- **Dimensions:** Graphene is a 2D nanomaterial; CNT is a 1D nanomaterial.
- **Properties:** While both have excellent electrical conductivity and strength, CNTs exhibit either metallic or semiconducting behavior depending on their chirality (how they are rolled). Graphene is a zero-bandgap semiconductor (semimetal).

Classification by Dimensions: (See 2083 Baishakh Q8 for 0D, 1D, 2D, 3D classification).

Question 2081 Ashwin Q8

Problem: What are nanofibers? Mention their applications in health and environment.

Solution: Nanofibers: (See definition in 2082 Bhadra Q8).

Applications:

- **Health/Biomedical:** Used as scaffolds for tissue engineering, as wound dressing materials to promote healing, and as drug delivery systems due to their high surface area and porosity.
- **Environment:** Used as highly efficient filter media for air purification (capturing PM2.5 and viruses) and water treatment (removing heavy metals, dyes, and bacteria) due to their tiny pore sizes and high permeability.

Chapter 7

Materials and Polymers

Question 2083 Baishakh & 2082 Bhadra Q9a

Problem: How does addition polymer differ from condensation polymer? Write down the mechanism of cationic polymerization reaction with a suitable example.

Solution: Differences:

- **Addition Polymerization:** Formed by the repeated addition of monomer molecules possessing double or triple bonds. No by-products are eliminated. The empirical formula of the polymer is identical to that of the monomer. Example: Polyethylene from ethene.
- **Condensation Polymerization:** Formed by step-growth reaction between monomers containing two or more reactive functional groups. Small molecules like H_2O , HCl , or NH_3 are eliminated as by-products. The empirical formula of the polymer is different from the monomer. Example: Nylon 6,6.

Mechanism of Cationic Polymerization: Initiated by an electrophile (Lewis acids like BF_3 , AlCl_3 along with a proton source like H_2O). *Example:* Polymerization of isobutylene to polyisobutylene.

1. **Initiation:** The catalyst reacts with the co-catalyst to generate a proton (H^+). The proton attacks the π electron cloud of the monomer to form a stable carbocation.
 $\text{H}_2\text{O} + \text{BF}_3 \longrightarrow \text{H}^+ + [\text{BF}_3\text{OH}]^-$
 $\text{H}^+ + \text{CH}_2=\text{C}(\text{CH}_3)_2 \longrightarrow \text{CH}_3-\text{C}^+(\text{CH}_3)_2$ (tert-butyl carbocation)
2. **Propagation:** The carbocation attacks another monomer molecule, transferring the positive charge to the end of the growing chain. $\text{CH}_3-\text{C}^+(\text{CH}_3)_2 + \text{CH}_2=\text{C}(\text{CH}_3)_2 \longrightarrow \text{CH}_3-\text{C}(\text{CH}_3)_2-\text{CH}_2-\text{C}^+(\text{CH}_3)_2 \dots \longrightarrow$ growing chain.
3. **Termination:** The reaction stops either by the combination of the growing macrocation with the counter-ion ($[\text{BF}_3\text{OH}]^-$) or by the elimination of a proton to form a terminal double bond.

Question 2083 Baishakh Q9b & 2081 Ashwin Q9

Problem: What is carbon fiber? Write the preparation and uses of Kevlar and polymethyl methacrylate (PMMA).

Solution: Carbon Fiber: It is a very strong, stiff, and lightweight fiber consisting almost entirely of carbon atoms arranged in long, thin crystalline structures. It is widely used in composites (Carbon Fiber Reinforced Plastics - CFRP).

Preparation and Uses of Kevlar: *Preparation:* Kevlar is an aramid polymer synthesized by the condensation polymerization of 1,4-phenylene-diamine (para-phenylenediamine) and terephthaloyl chloride in an amide solvent. $n \text{H}_2\text{N}-\text{C}_6\text{H}_4-\text{NH}_2 + n \text{ClOC}-\text{C}_6\text{H}_4-\text{COCl} \longrightarrow [-\text{NH}-\text{C}_6\text{H}_4-\text{NH}-\text{CO}-\text{C}_6\text{H}_4-\text{CO}-]_n + 2n \text{HCl}$ *Uses:* Bulletproof vests, combat helmets, high-performance ropes, and cut-resistant gloves.

Preparation and Uses of PMMA (Polymethyl methacrylate): *Preparation:* Prepared by the free-radical addition polymerization of methyl methacrylate in the presence of organic peroxides as initiators. $n \text{CH}_2=\text{C}(\text{CH}_3)\text{COOCH}_3 \longrightarrow [-\text{CH}_2-\text{C}(\text{CH}_3)(\text{COOCH}_3)-]_n$ *Uses:* Used as a shatter-resistant glass substitute (Plexiglas, Lucite) in aircraft windows, optical lenses, and illuminated signs.

Question 2082 Bhadra Q9a

Problem: Define fiber reinforced polymer. Write the preparation and use of polycarbonate and epoxy resin.

Solution: Fiber Reinforced Polymer (FRP): A composite material consisting of a polymer matrix reinforced with fibers (like glass, carbon, or aramid). The matrix binds the fibers together, while the fibers provide exceptional tensile strength and stiffness.

Polycarbonate: *Preparation:* Synthesized by the condensation polymerization of Bisphenol-A and phosgene (COCl_2) in the presence of a base (like NaOH). $n \text{HO}-\text{C}_6\text{H}_4-\text{C}(\text{CH}_3)_2-\text{C}_6\text{H}_4-\text{OH} + n \text{COCl}_2 \longrightarrow [-\text{O}-\text{C}_6\text{H}_4-\text{C}(\text{CH}_3)_2-\text{C}_6\text{H}_4-\text{O}-\text{CO}-]_n + 2n \text{HCl}$ *Uses:* Bulletproof glass, safety goggles, CD/DVDs, and electronic casings.

Epoxy Resin: *Preparation:* Formed by the step-growth polymerization of Bisphenol-A and epichlorohydrin in the presence of NaOH to form a prepolymer, which is then cross-linked ("cured") using a hardener (a polyamine). *Uses:* High-performance adhesives, coatings for electronic components, and matrix for FRPs (like carbon fiber composites).

Question 2082 Baishakh Q9b

Problem: Discuss about an addition polymerization with an example. Explain about the free radical mechanism of Polymerization.

Solution: Addition Polymerization: (See definition above. Example: Ethene $\rightarrow$ Polyethylene).

Free Radical Mechanism: It involves three main steps and is initiated by a substance that readily cleaves to form free radicals (like benzoyl peroxide). Let $\text{R}^\bullet$ be the radical and $\text{CH}_2=\text{CH}-\text{Y}$ be the monomer.

1. **Initiation:** The initiator decomposes to form free radicals. The radical attacks the double bond of the monomer, generating a new radical species. $\text{R}^\bullet + \text{CH}_2=\text{CH}-\text{Y} \longrightarrow \text{R}-\text{CH}_2-\text{CH}^\bullet-\text{Y}$
2. **Propagation:** The new radical attacks another monomer molecule, transferring the

radical to the end of the chain. This process repeats rapidly, building the polymer chain. $R-CH_2-CH^\bullet-Y + CH_2=CH-Y \longrightarrow R-CH_2-CH(Y)-CH_2-CH^\bullet-Y$

3. **Termination:** The growth stops when two growing radical chains combine (coupling) or when one transfers a hydrogen atom to another, forming one saturated and one unsaturated polymer chain (disproportionation).

Question Short Notes

Problem: Write short notes on: Chitin, Polythiazyl, Cellulose, Polyphosphazene, Polyaniline (PANI), Collagen, and Hydration of Cement

Solution:

- **Chitin:** A natural biopolymer, it is a long-chain structural polysaccharide of N-acetylglucosamine. It is the main component of the exoskeletons of arthropods (insects, crustaceans) and fungal cell walls. It is highly biocompatible and used in surgical threads.
- **Polythiazyl ((SN)_x):** An inorganic, synthetic polymer consisting of alternating sulfur and nitrogen atoms. It was the first non-metallic polymer discovered to exhibit metallic electrical conductivity and becomes a superconductor at very low temperatures.
- **Cellulose:** A naturally occurring organic polymer (a polysaccharide) consisting of a linear chain of several hundred to many thousands of $\beta(1 \rightarrow 4)$ linked D-glucose units. It is the primary structural component of the cell wall of green plants (e.g., cotton, wood).
- **Polyphosphazene:** A class of inorganic-organic polymers with a backbone of alternating phosphorus and nitrogen atoms, with two side groups attached to each phosphorus. They are flame retardant, highly flexible at low temperatures, and used in biomedical materials and advanced elastomers.
- **Polyaniline (PANI):** A highly important conducting polymer (an intrinsically conducting polymer). It exists in various oxidation states; the "emeraldine salt" form is highly conductive. Synthesized by oxidative polymerization of aniline. Used in anti-static coatings, sensors, and supercapacitors.
- **Collagen:** The most abundant structural protein (a natural polymer) in the extracellular matrix of various connective tissues in animals. It forms a characteristic triple helix structure. It is used widely in biomedical applications, cosmetics, and as gelatin when denatured.
- **Hydration of Cement:** When water is added to Portland cement, a series of complex exothermic chemical reactions occur. The main compounds, tricalcium silicate (C3S) and dicalcium silicate (C2S), react with water to form a rigid, porous interlocking gel of Calcium Silicate Hydrate (C-S-H) and calcium hydroxide (Ca(OH)₂), which gives cement its strength.

Chapter 8

Explosives and Lubricants

Question 2083 Baishakh Q10

Problem: What are plastic explosives? Write the preparation and application of nitroglycerine.

Solution: Plastic Explosives: These are a form of high-brisance explosives that are soft, moldable, and dough-like in consistency. They consist of a primary explosive chemical (like RDX or PETN) mixed with a plasticizer, a polymeric binder, and sometimes oil. Because they are moldable, they can be easily shaped to direct the blast energy (shaped charges) and are relatively stable against shock. Example: C-4.

Preparation of Nitroglycerine (TNG / GTN): It is prepared by the nitration of glycerol (glycerin) using a cold nitrating mixture (concentrated HNO_3 and concentrated H_2SO_4) under strict temperature control ($< 20^\circ\text{C}$).
$$\text{CH}_2(\text{OH}) - \text{CH}(\text{OH}) - \text{CH}_2(\text{OH}) + 3\text{HNO}_3 \xrightarrow[20^\circ\text{C}]{\text{H}_2\text{SO}_4} \text{CH}_2(\text{ONO}_2) - \text{CH}(\text{ONO}_2) - \text{CH}_2(\text{ONO}_2) + 3\text{H}_2\text{O}$$

The product is an oily, highly shock-sensitive liquid.

Applications: - **Explosives:** Mixed with kieselguhr to form Dynamite (rendering it safer to handle). - **Medical:** Used in very small doses as a vasodilator to treat angina pectoris (chest pain) in heart disease.

Question 2082 Bhadra Q10

Problem: What is primary explosive? Write the preparation and uses of TNT.

Solution: Primary Explosive: A highly sensitive explosive that easily detonates when subjected to heat, friction, static electricity, or mild mechanical shock. Because of their sensitivity, they are used in small quantities in detonators or blasting caps to trigger a larger, less sensitive secondary explosive. Example: Lead azide, Mercury fulminate.

Preparation of TNT (Trinitrotoluene): TNT is prepared by the stepwise nitration of toluene using a nitrating mixture (conc. HNO_3 and conc. H_2SO_4) at elevated temperatures. The reaction goes through mono- and di-nitrotoluene before forming 2,4,6-trinitrotoluene.
$$\text{C}_6\text{H}_5\text{CH}_3 + 3\text{HNO}_3 \xrightarrow{\text{H}_2\text{SO}_4, \Delta} \text{C}_6\text{H}_2(\text{CH}_3)(\text{NO}_2)_3 + 3\text{H}_2\text{O}$$

Uses of TNT: - Widely used as a secondary military explosive in shells, bombs, and grenades

because of its insensitivity to shock and friction, allowing safe handling and firing. - Used in industrial and mining blasting.

Question 2082 Baishakh Q10a

Problem: What are solid lubricants? Why is graphite used as a lubricant?

Solution: Solid Lubricants: These are solid materials used to reduce friction and wear between two surfaces sliding against each other, especially under extreme conditions (high temperature, high vacuum, high load) where liquid lubricants would vaporize, freeze, or be squeezed out. Examples: Graphite, Molybdenum disulfide (MoS_2), PTFE.

Why Graphite is used: Graphite consists of layers of carbon atoms arranged in hexagonal rings. Within a layer, atoms are tightly bonded by strong covalent bonds. However, the separate layers are held together by weak Van der Waals forces. This allows the layers to easily slide over one another when shear force is applied, providing excellent lubricating properties.

Question 2081 Ashwin Q10

Problem: What are the requisites of a good paint? Discuss about the impact of paint on a human health.

Solution: Requisites of a Good Paint: 1. It should have high hiding power (opacity) to cover the surface well. 2. It should dry within a reasonable time and form a smooth, uniform, and adherent film. 3. The resulting film should be tough, durable, and resistant to weathering, water, and corrosion. 4. The color should be stable and not fade upon exposure to sunlight. 5. It should not crack, peel, or blister easily.

Impact of Paint on Human Health: - **VOCs (Volatile Organic Compounds):** Many oil-based paints release VOCs (like benzene, toluene) as they dry. Inhaling these can cause headaches, dizziness, respiratory irritation, and long-term exposure may damage the liver, kidneys, or central nervous system. - **Heavy Metals:** Older paints or specialized anti-corrosive paints may contain heavy metals (like Lead or Chromium). Lead exposure (e.g., from peeling old paint) can cause severe neurological damage, particularly in children. - **Sensitization:** Some components (like epoxies or polyurethanes) can cause skin sensitization, allergic contact dermatitis, or asthma upon repeated exposure.

Question General

Problem: Write the types and functions of lubricants.

Solution: Types of Lubricants:

- **Liquid Lubricants:** Mineral oils, synthetic oils (e.g., silicone oils), vegetable oils.
- **Semi-solid Lubricants:** Greases (oil mixed with a thickener like soap).
- **Solid Lubricants:** Graphite, MoS_2 , PTFE.

Functions of a Lubricant: 1. **Friction Reduction:** Forms a film between moving parts to prevent metal-to-metal contact, reducing friction and wear. 2. **Cooling Effect:** Absorbs and carries away heat generated by friction (especially liquid lubricants). 3. **Protection:** Protects metal surfaces against rust and corrosion. 4. **Sealing:** Greases can act as a seal against dust, dirt, and moisture. 5. **Cleaning:** Carries away wear particles and contaminants from the contact zone.